changed?

6.4.2 Bayesian optimization for hyperparameter tuning

This exercise, implemented in `CH06/03 - Exercise 2.ipynb` applies BayesOpt to an objective function that simulates the accuracy surface of a support-vector machine model in a hyperparameter tuning task. The x-axis denotes the value of the penalty parameter C, while the y-axis denotes the value for the RBF kernel parameter γ. See the exercises in Chapter 3 and Chapter 4 for more detail. There are multiple steps:

1. Recreate the Bayesian optimization loop in `CH07/03 - Exercise 2.ipynb`, including the outer loop that implements repeated experiments.

   SIDEBAR **Repeated experiments in Bayesian optimization**

   Refer to Step 9 of Exercise 2 from Chapter 4 to see how we can run repeated experiments in Bayesian optimization.

2. Run the Max-value Entropy Search (MES) policy. Since our objective function is two-dimensional, we should increase the size of the Sobol sequence used by MES. For example, you can set it at 2000. Observe its aggregated performance.

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